

1 Introduction To Communication System

1.1 Communication

Q. Define noise, distortion and interference in the communication system. Describe the types and causes of internal and external noise that may affect the communication system.

Q. Define Any 10 Communication Impairments.

Q. Compare digital and analog communication systems.

Q. Draw the generic block diagram of Analog communication system.

Q. Draw the generic block diagram of Digital communication system.

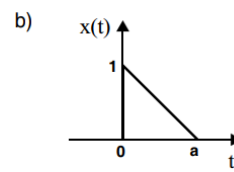
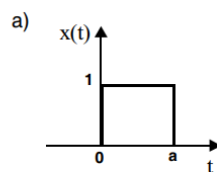
Q. Calculate the theoretical channel capacity of a communication link if its bandwidth is 4000 Hz and its operating SNR is 20 dB.

Q. Write About Transmission Mediums .

1.2 Signal And System

Example 2.6

Sketch the even and odd parts of the following signals.



Solution

a) The even part of the signal is given by, $x_e(t) = \frac{1}{2}[x(t) + x(-t)] = \frac{1}{2} x(t) + \frac{1}{2} x(-t)$ (1)

The odd part of the signal is given by, $x_o(t) = \frac{1}{2}[x(t) - x(-t)] = \frac{1}{2} x(t) - \frac{1}{2} x(-t)$ (2)

Example 2.7

Construct the block diagram and signal flow graph of the system described by the equation,

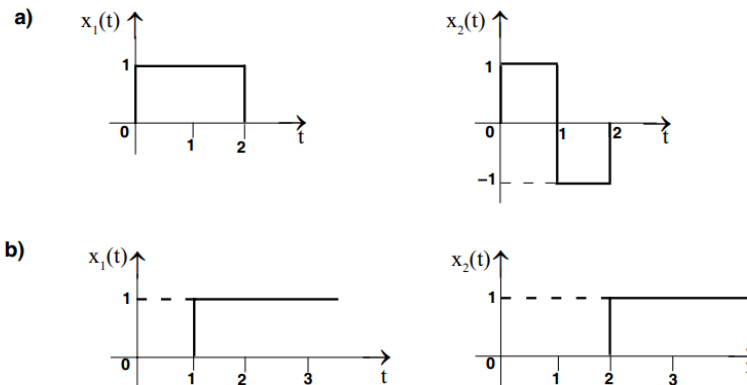
$$\frac{d^2y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 3 y(t) = 4 \frac{dx(t)}{dt} + 5 x(t)$$

Q. Signal flow using 5 mathematical operations. [Exam Pyls]

18. Prepare a signal flow diagram of a system that combines at least seven mathematical operations in an equation. [2081 Bhadra, 2082 Baishakh — CT602]

Example 2.23

Perform convolution of the following signals, by graphical method and sketch the resultant signal.



26. Find the odd and even symmetry of the given signals graphically (triangular and right-triangular pulses of amplitude 2 spanning $t = 0$ to 2).

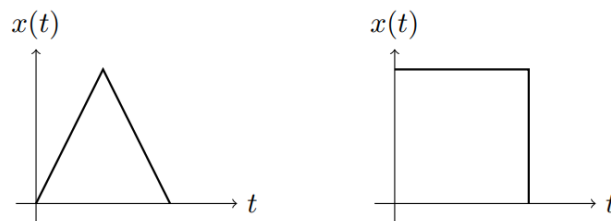


Figure 1: *
Signals for odd/even symmetry evaluation (reconstructed)

[2083 Baishakh — CT602]

27. With suitable examples, explain how time invariance of a system is checked. [2082 Bhadra — EX656]
28. With suitable examples, explain how linearity of a system is checked. [2083 Baishakh — EX656]
29. Mention the conditions and properties of Fourier transformation. [2083 Baishakh — EX656]
30. Plot the magnitude and phase spectrums of $x(t) = 6 - 8 \sin(2\pi 160t + 30^\circ) - 6 \cos(\pi 160t - 50^\circ) + 2 \sin(300\pi t + 120^\circ)$. [2082 Bhadra — ENCT253]

36. Define LTI system. Convolve the following signals graphically:

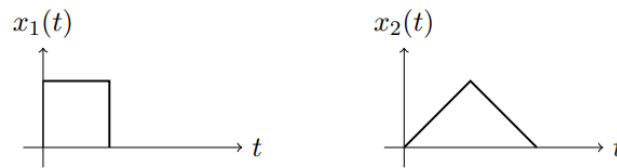
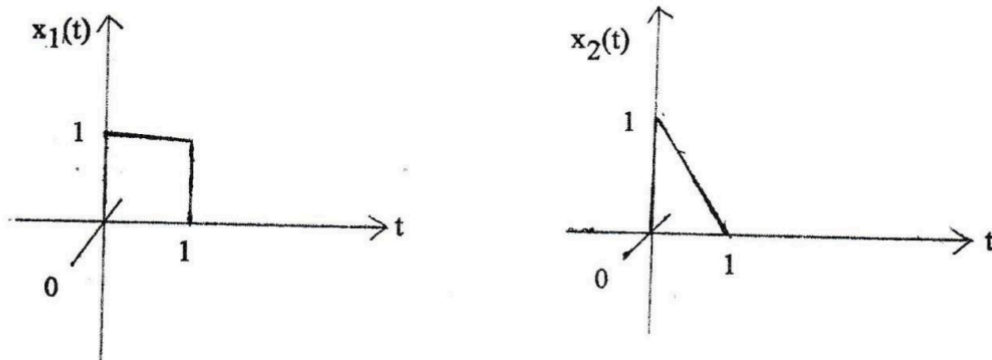


Figure 3: *
Rectangular and triangular pulse to be convolved graphically

[2080 Bhadra — CT602]

[2080 Bhadra | Regular]

2. Define a causal system and a stable system with example . Evaluate power and energy of a signal given as $x(t) = e^{-2|t|}$. Justify whether it is energy signal or power signal. [4+6]
3. Define LTI system. Convolve the following signals graphically. [2+6]



[2081 Baishakh | Back]

2. Define energy and power signal with examples. Check whether the following system is linear, causal and time invariant or not. [4+6]
 $y(t) = \log x(t)$
3. State the properties of Fourier Transform. Convolve the following signals graphically. [2+6]



2 Amplitude Modulation

[Exam] Explain AM

Q. How to generate AM signal with circuit. [Exam]

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Q. AM is very noisy. give ans.

- A काई उत्तर [Exam]
- Swing of A $[A_c + m(t)] \sin(\omega_c t)$

Q. Describe Hilbert Transformation and its properties .

3 Angle Modulation

Q. Describe the working of FM.

Q. Write Short Notes On

- a. pre-emphasis and de-emphasis network
- b. AM Vs FM
- c. Why FM is noise immune .
- d. Armstrong's method for FM

4 Pulse Modulation

Q. Explain the working principle of PCM with necessary figures and equations.

5 Channel Bandwidth Utilisation

Q. What is frequency division multiplexing (FDM)? Explain FDM hierarchy used in telephony.

Q. What is time division multiplexing (TDM)? Explain TDM hierarchy used in telephony.

Q. Why statistical TDM is preferred than synchronous TDM?

Q. Write short notes on:

a. E1 and T1 Hierarchies

b. CDMA

Q. Differentiate between FDMA and TDMA.

Q. Compare packet vs circuit switching .

Q. Explain DSSS (Direct-Sequence Spread Spectrum).

6 Baseband Digital Data Transmission

Q. Any 12 digital bitstream encoding using all kinds of line coding mentioned in the syllabus .

Example :

Q. Represent 100111010 using following encoders: (a) Polar RZ (b) Bipolar NRZ (c) AMI (d) Manchester. [2078 Chaitra]

Q. Given the binary sequence 1101010111 represent in Unipolar RZ, Bipolar NRZ, Polar NRZ and Manchester encoders. Explain communication impairments with examples. [2079 Chaitra]

Q. Given the binary sequence 1001101101, represent in unipolar NRZ, polar RZ, AMI, and Manchester codes. [2081 Bhadra]

Q. Represent the given binary sequence 110011000001 in unipolar RZ, Manchester and HDB3 encoders. [2080 Chaitra]

Q. Represent 100111010 using Unipolar RZ, Bipolar NRZ, Manchester and AMI encoding technique. [2080 Ashwin]

Q. Encode the given bit sequence 101000001 using Unipolar RZ, AMI and HDB3 encoders. [2081 Ashwin]

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Q. Explain Shannon-Hartley channel capacity theorem and its limitations.

Q. Differentiate between message and information. Relate the message, the entropy and the information.

Q. Encode Following Using Huffman And Shannon-Fano Coding .

1. Tara Baji Lai Lai
2. Phoola Ko Thunga Bahera Gayo Ganga Ko Panima
3. Engineer Ko Betha Aru Lai Ke Thaha
4. Kun Mandir Ma Janchhau Yatri
5. Rain Rain Go To Spain
6. Any Signal And System Related Formula
7. Ma Mare Pani Mero Desh Bachi Rahos
8. Mississippi is missing
9. See eye-to-eye (LOS)

7 Digital Modulation Techniques

Q. Describe ASK, PSK and FSK with signal diagrams.

Q. Explain working of QAM .

Q. Draw Constellation diagram for n - QAM . for $n = 8, 16, 32$

Q. Explain working of Minimum Shift Keying (MSK) and GMSK .

Q. What is eye diagram.

8 Error Detection and Correction Coding

53. What is code vector? Visualize a 3-bit codeword as code vector.

-Visualization of 'n' bit codeword in an n-dimensional space as a vector whose elements or coordinates are the bits in the codeword.

54. What are linear block codes?

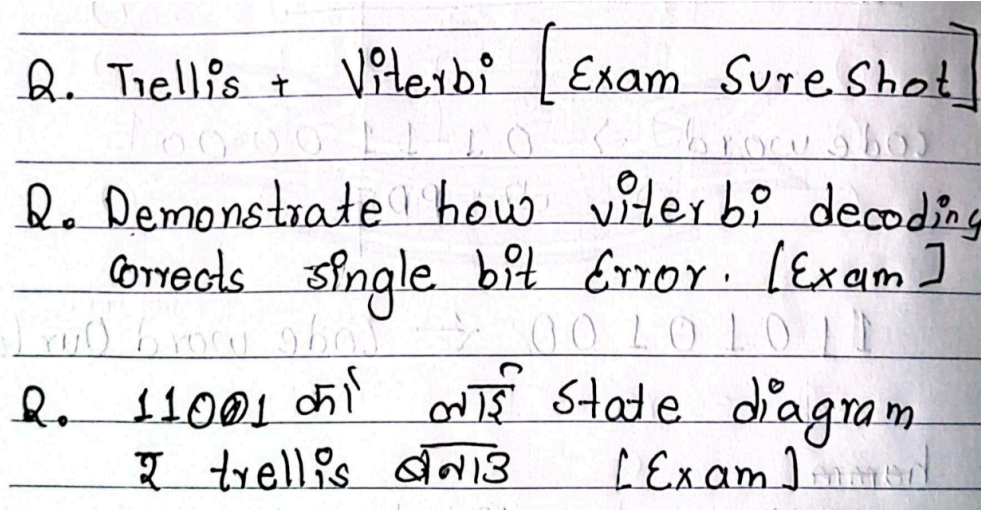
-For the block of 'k' message bits, (n-k) redundant check bits are added. This gives (n,k) block codes with 'n' bits in the o/p of the channel encoder

-Block codes are known as linear if the mod-2 addition of any two code vectors produces another code vector.

271. For a (6,3) code, the generator matrix is

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

Find (i) All corresponding code vectors (ii) Minimum Hamming distance (iii) Verify that this code is a single error correcting code (iv) Parity check matrix (v) Determine transmitted codeword if received code is 100011. [2079 Bhadra — CT602]



Q. Construct a (7, 4) binary CRC using a generator polynomial $g(x) = x^3 + x^2 + 1$ with data Vector (1011).

Q. Demonstrate how LBC corrects a single bit error.

Q. Explain working of CRC-4 .

9 Noise in Communication Systems

Q. Calculate Ideal low-pass and RC filtering of white noise.

Q. Write Short Notes About

- a. Noise equivalent bandwidth of a filter
- b. PSDF of white noise

Q. Explain Optimum detection of a pulse in additive white noise.

Q. Compare Error Probabilities Of Different Digital Modulation Scheme .